MPLS JAPAN

# Front-haul networking for 5G: An analysis of technologies and standardization

www.huawei.com

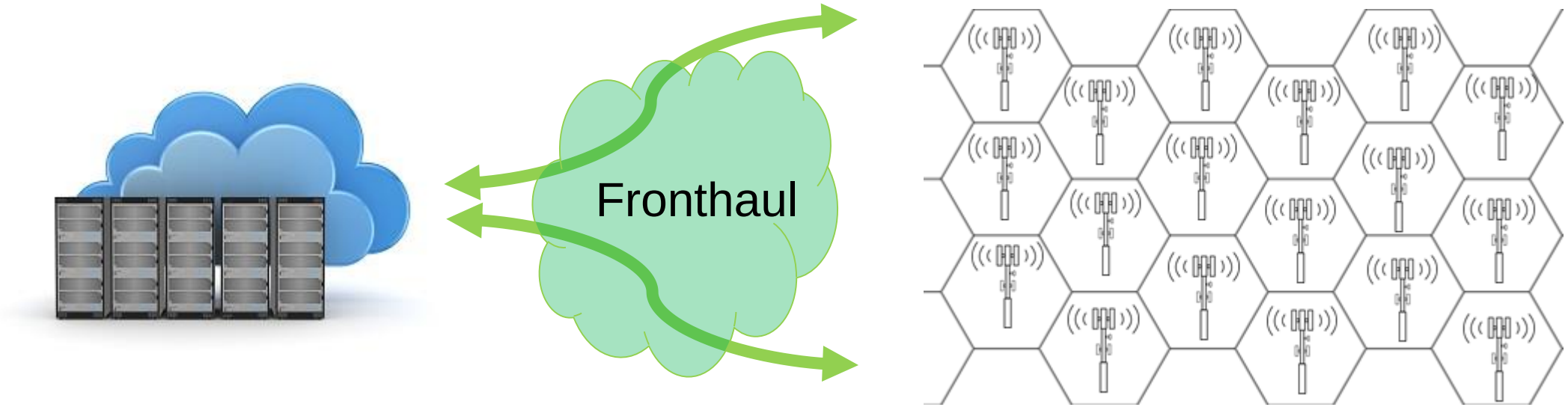
Hassan Halabian
hassan.halabian@Huawei.com
Canada Research Center
November 2017

HUAWEI TECHNOLOGIES CO., LTD.

HUAWEI

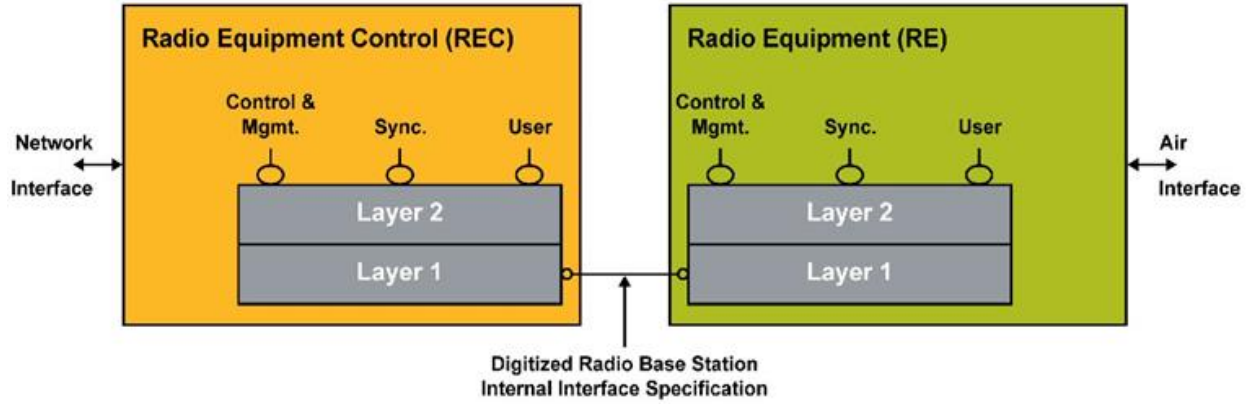
# Cloud-RAN

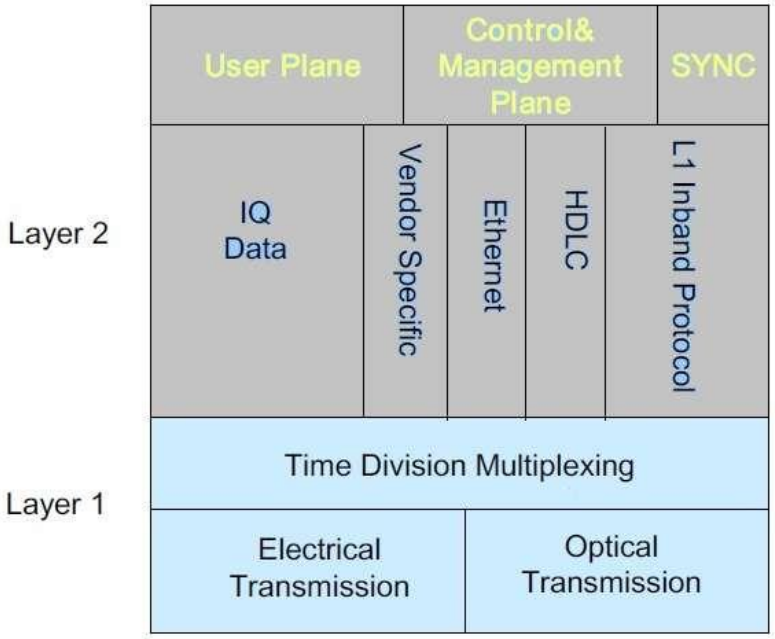
- **Advantages:**
  - Simplify the RRH site – Full outdoor, low cost
  - Better RAN network performance - Facilitates cooperative radio schemes, e.g., ICIC, CoMP, Massive MIMO
  - Future oriented – RAN virtualization



# CPRI widely Deployed For 4G

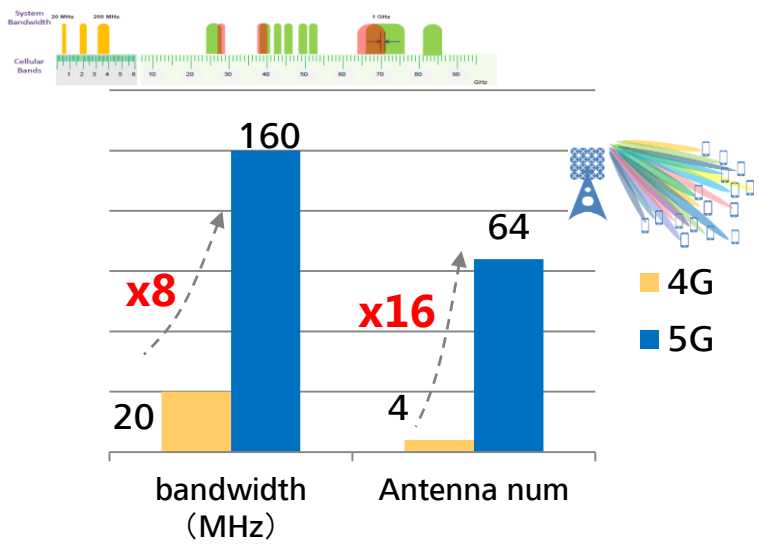
- CPRI is - TDM based synchronous interface/ Symmetrical serial data link/Steady data stream, ”Always on”
- Embedded Ethernet or HDLC streams (C&M plane data)
- I/Q data is the time samples of the wireless analog signal (Data plane)
- **Front-haul capacity linearly scales with**
  - Number of antenna ports/Number of sectors
  - Sampling rate (Radio spectrum BW)/I/Q bit-width
  - Transmission overhead (coding and headers, etc.)/compression factor etc
- BER=$10^{-12}$



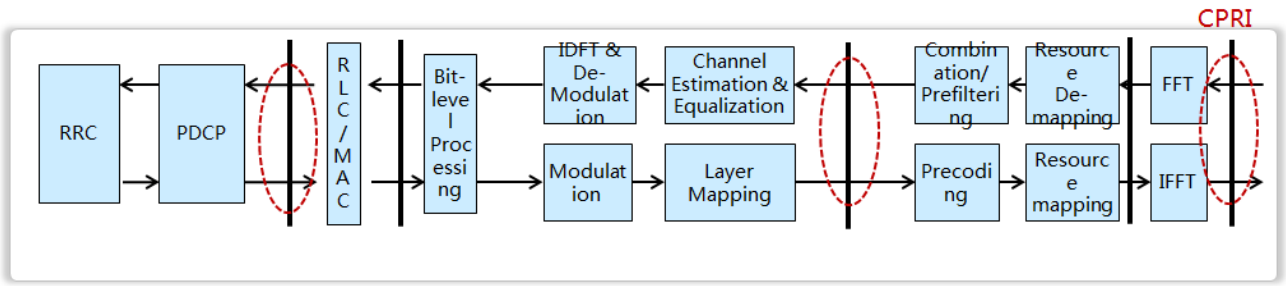



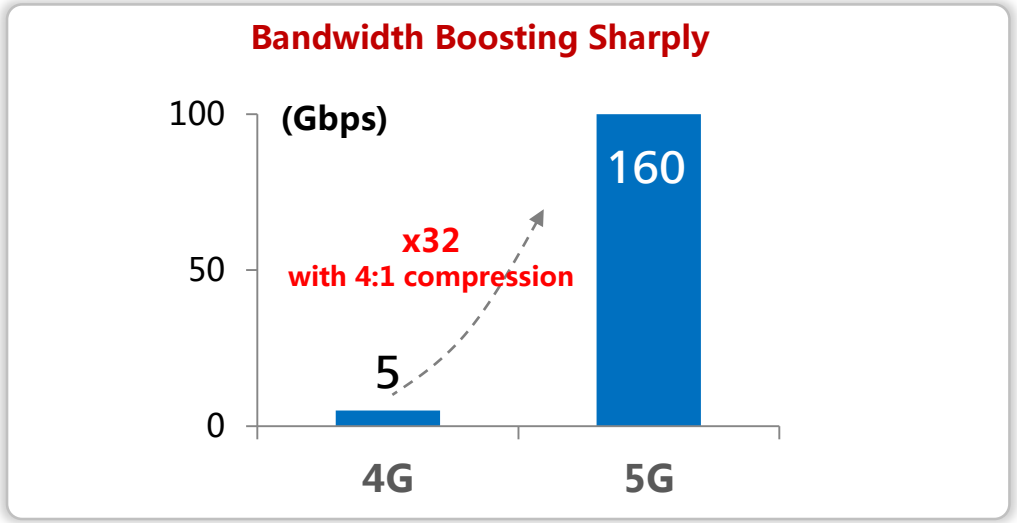
# If Use 4G RAN Architecture for 5G

- **Facing high bandwidth challenge**
  - Big challenge for Massive MIMO (64T/64R) – Beyond 100Gbps requirement
  - But the real fronthaul bandwidth depends on configuration – air spectrum bandwidth, Massive MIMO types etc



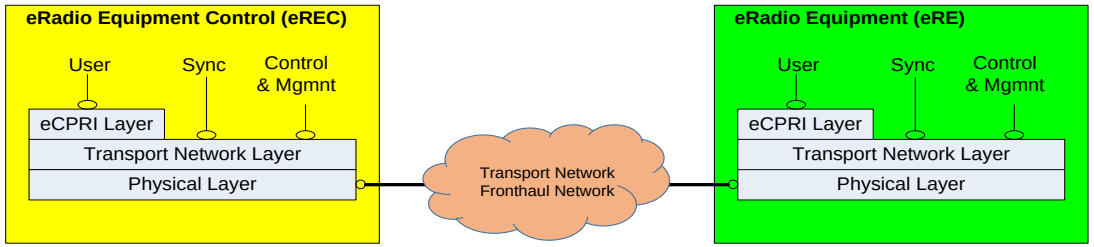

CPRI Split Point
50~130 us

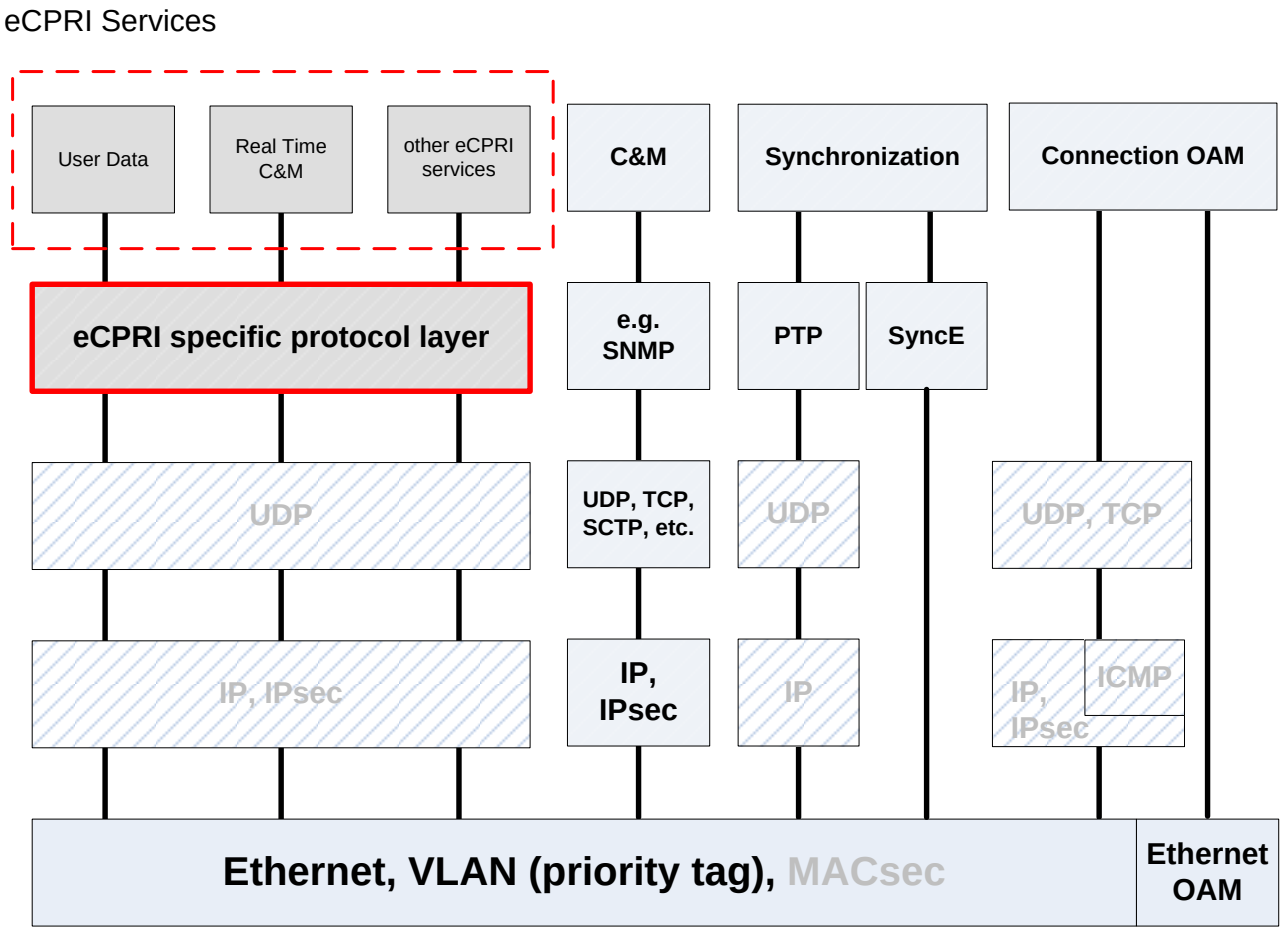


# eCPRI Protocol Stack

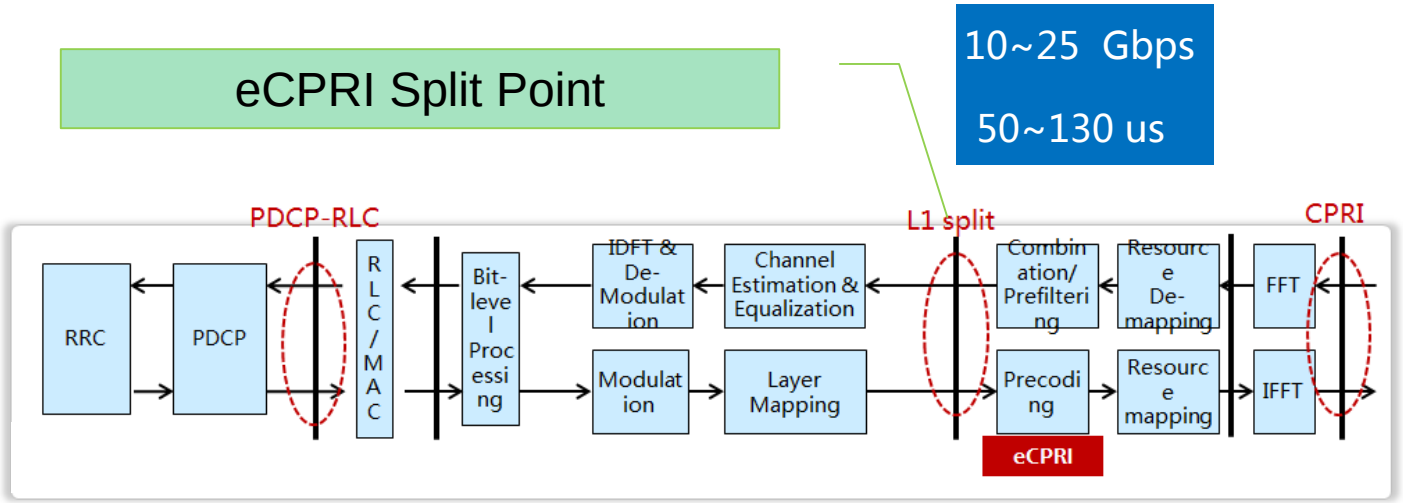
- **eCPRI Laye1:**
  - eCPRI over electrical cable
  - eCPRI over electrical backplane
  - eCPRI over optical fiber
- **eCPRI over Ethernet:**
  - eCPRI messages shall be transmitted in standard Ethernet frames
- **eCPRI over IP:**
  - eCPRI messages shall be transmitted in UDP/IP packets



# If Use New RAN Architecture for 5G

- **When compare CPRI with eCPRI, we found more challenges**







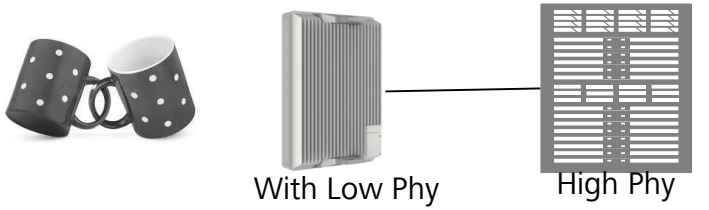
•Ignore RAN challenges
•eCPRI used for fronthaul

## 1: 5G AAU weight increasing



Compared with CPRI part Phy function embedded in 5G AAU

## 2: 5G AAU upgrade when add new features




## 3: Do not forget 4G: CPRI + eCPRI

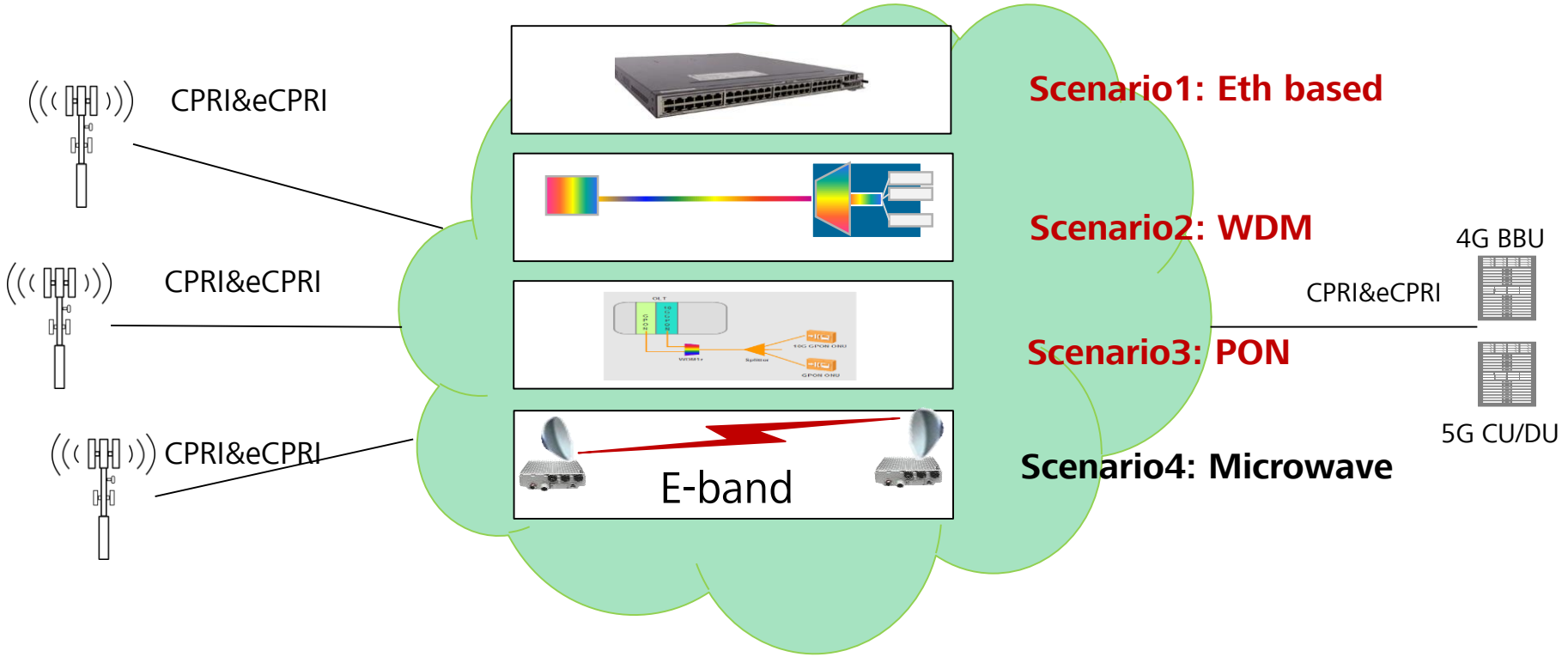
# Possible 5G Fronthaul Solutions

- **Increasing 5G site number faces fiber limitation**
- **Possible transport solutions:**

Site number increasing due to 4.5GHz/28GHz



Limited fiber resource



# Eth based Fronthaul Research Introduction

## Fronthaul Requirements Side

RAN Requirements

3GPP A GLOBAL INITIATIVE

Define RRH/BBU Interface Implementation

CPRI Common Public Radio Interface

ERICSSON HUAWEI NEC NOKIA

## Ethernet Based Fronthaul Network

IEEE 802.1 time sensitive networks

Last PAR 2018/07, SA Revcom approval: 2019/05
802.3br and 802.1Qbu : Frame preemption
802.1Qbv – Traffic Scheduling (time-gated queues)

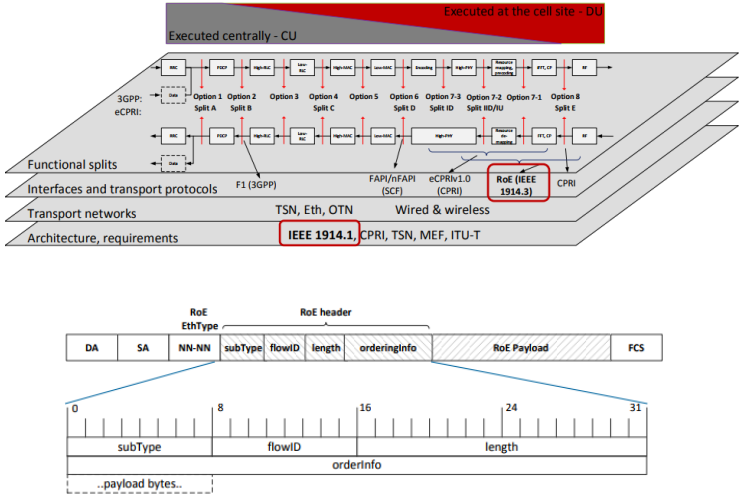



- Current focus: Profile(s) for current (CPRI 7.0)
- Next step is eCPRI
- Joint effort with CPRI Cooperation
- Joint effort with IETF DetNet WG

CPRI Common Public Radio Interface

IETF DetNet WG

IEEE 1914 Working Group







- Another Team for Fronthaul based on Eth
- Already demo for CPRI over Eth
- Define new Eth Type for Fronthaul

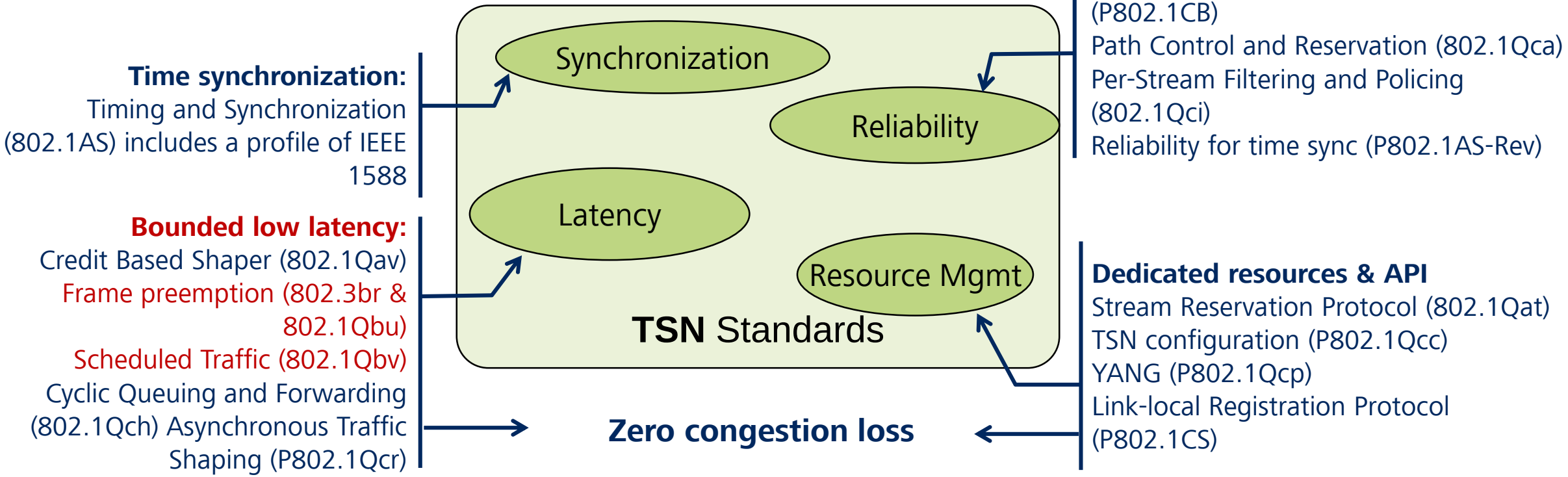
# IEEE 1914.3 – CPRI over Ethernet

- **Standard for Radio Over Ethernet (RoE) Encapsulations and Mappings, defines**
  - The encapsulation of digitized radio
  - mapper/demapper functions to transfer CPRI streams over Ethernet.
  - The header format for encapsulation of existing digitized radio transport formats.
- **Three types of encapsulation techniques**
  - structure–agnostic mapper: is only a container for the encapsulated digitized radio transport frames.
  - structure-aware mapper: has detailed knowledge of the encapsulated digitized radio transport format content
  - native RoE packet.

| | 1914.3 | eCPRI |
|---|---|---|
| Technology | Only Ethernet | UDP/IP/Eth and Eth. - any packet transport |
| Bandwidth | CBR (CPRI encapsulation) and VBR (Native RoE only) | User traffic related, VBR( ~10 fold reduction) |
| Topology | P2P connection, P2P/Chain/Ring | P2MP connection, flexible topology |
| RAN | IQ time domain and freq. domain data (Native RoE only) | Any functional split |
| Proprietary Performance Enhancement Support | Limited | Yes |

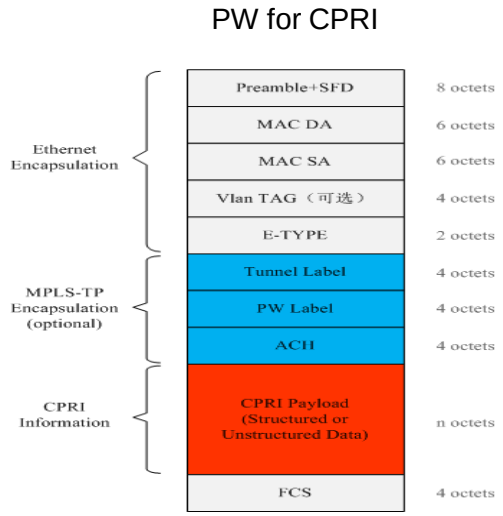
# IEEE 802.1 TSN Technology Overview

- Converged network for Information Technology (IT) and Operational Technology (OT) in LAN
- Multi profiles for different applications – Video, Factory, Telecom…

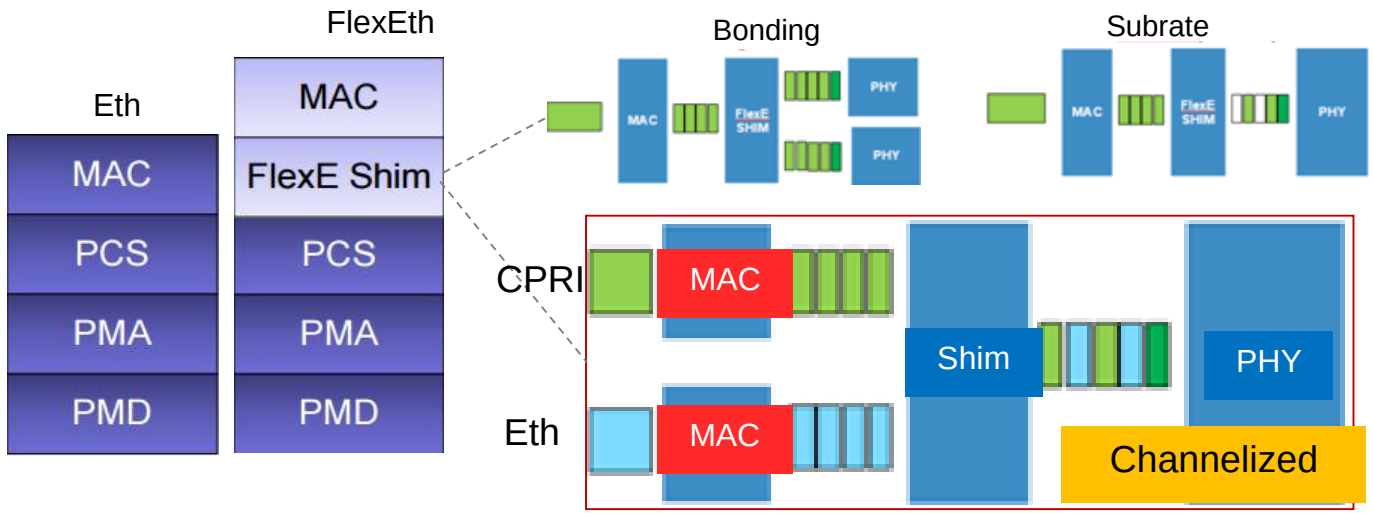


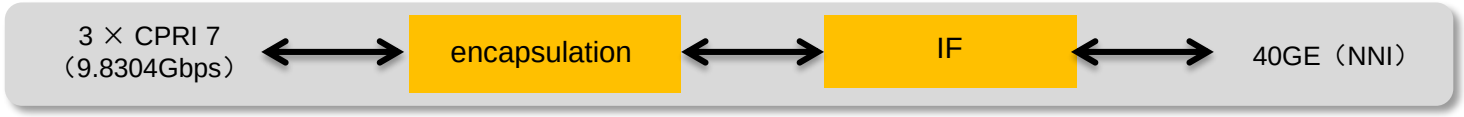
# Eth based Solution – Prototype Test

❑ Encapsulation of CPRI frame



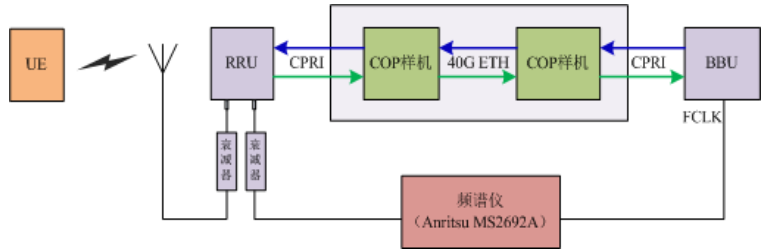

❑ Multiplexing with FlexE





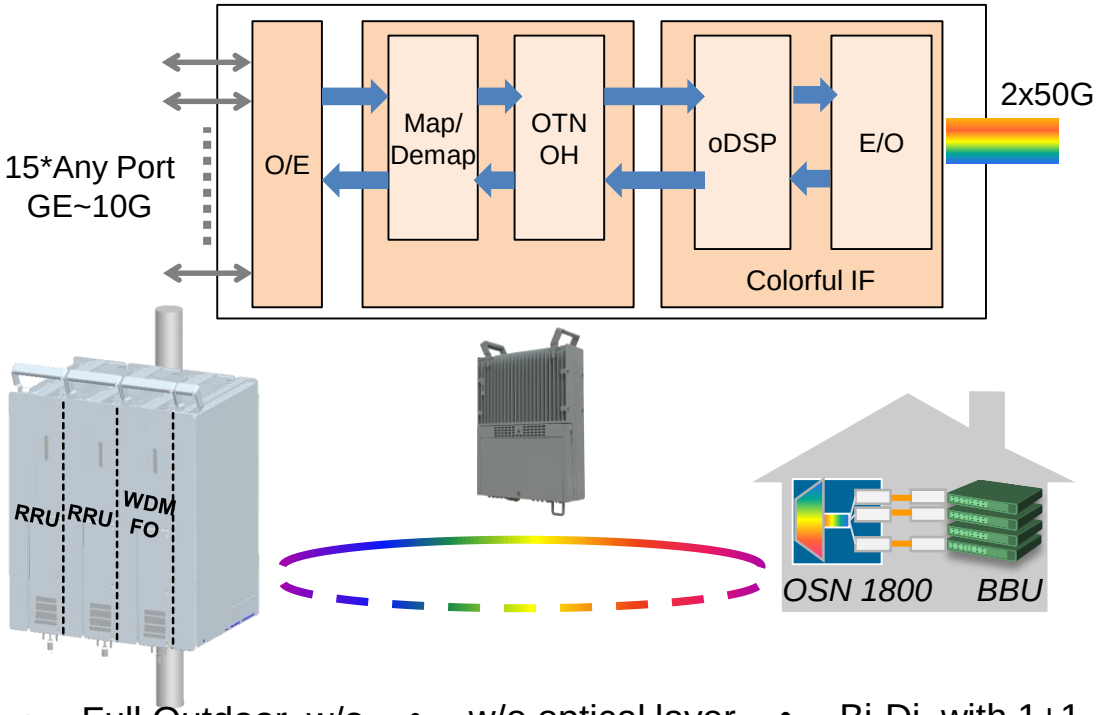

FPGA based, only lab POC




- **2016 – Lab POC, CPRI over Eth via PW(same as E1 PW), FlexEth;**
- **No bandwidth saving, device needed when CPRI and eCPRI at the same site**

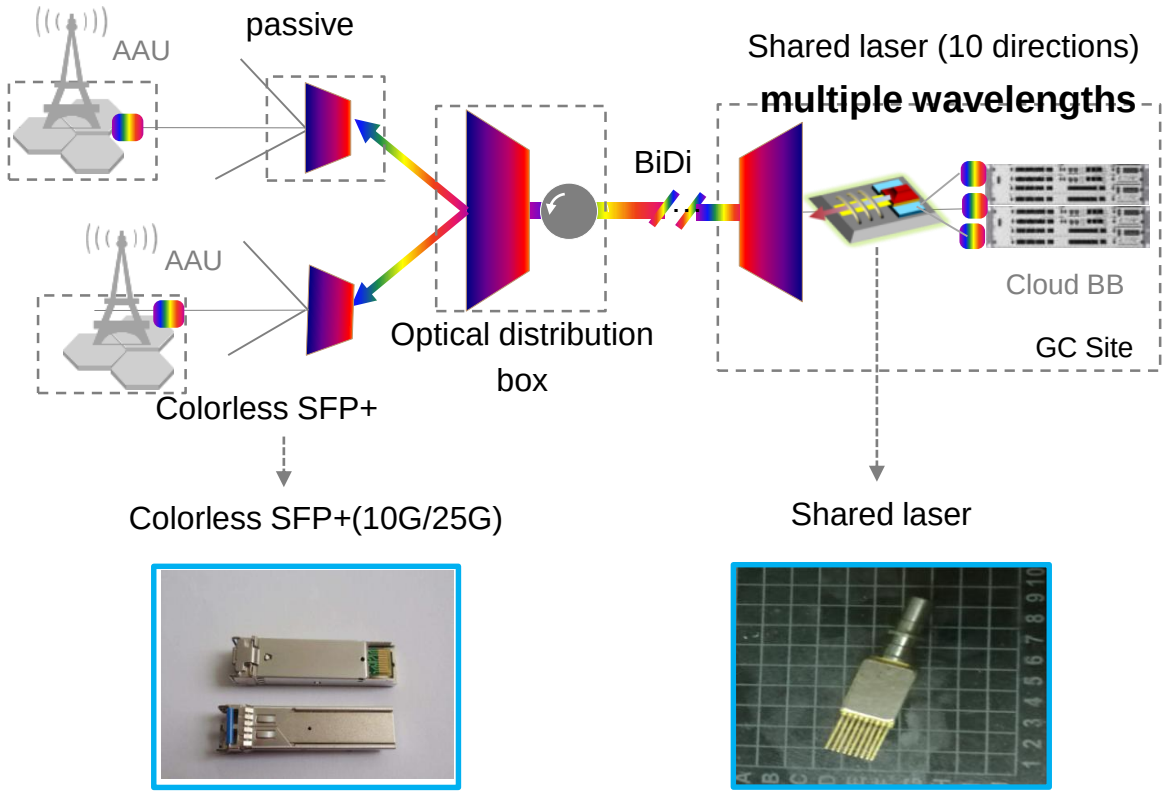
# WDM Based Solution – FO/Colorless Module

## Current FO Solution



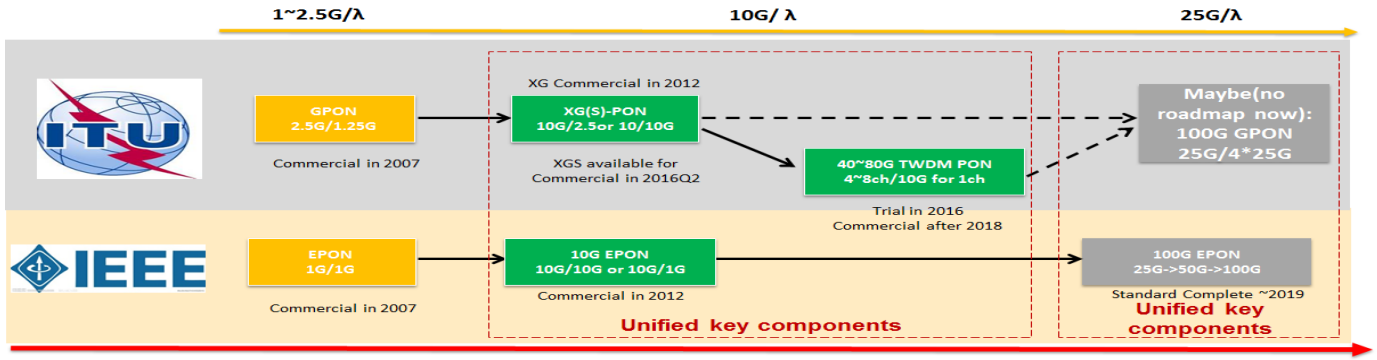

- Full Outdoor, w/o site configuration
- w/o optical layer commissioning
- Bi-Di, with 1+1 protection

## Centralized WDM Solution(Under R&D)

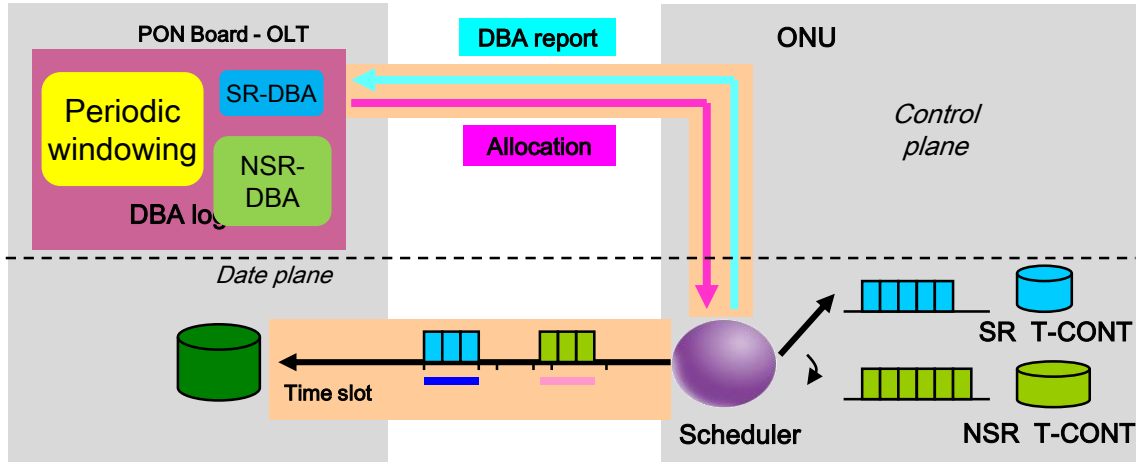



- **Compatible with CPRI/eCPRI – transparent for any traffic**
- **Simple, low power consumption and save site fibers – without device**

# PON Based Solution— Optimized DBA Algorithm

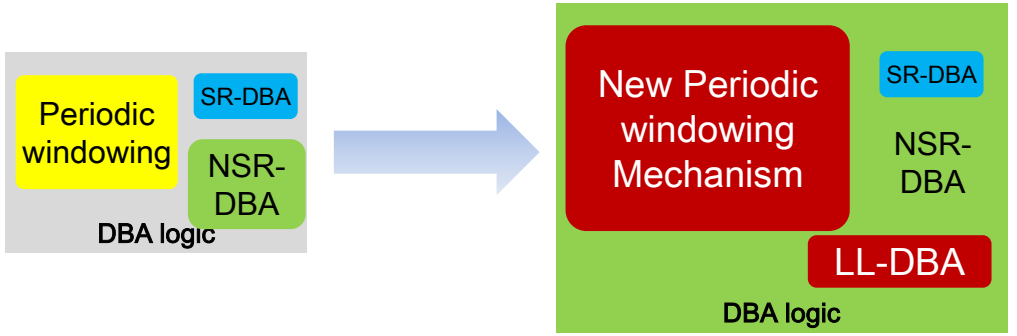



## Current NG-PON DBA ：350~550us




- Assured-DBA/Max DBA, status report before bandwidth allocation : 1-2ms.
- Fixed-DBA, no need status report, delay will be 100-200us
- Periodic windowing, add delay 200-300us

## DBA Optimization for 5G：50us~100us




- LL-DBA, low latency DBA in future, delay will be about 50us~100us.
- Periodic windowing should have new mechanism to eliminate latency to 0.

**Notes: chipset and hardware must be changed to support 50us~100us low latency DBA**

# Fronthaul Solution Summary

| 1<br>WDM - Colorless | 2<br>Eth – RoE | 3<br>IP/Eth – eCPRI | 4<br>PON |
|---|---|---|---|
| • Transparent for CPRI/eCPRI: 4G&5G Compatible<br>• No site device: Simple, low power consumption<br>• Innovation: new optical module | • CPRI mapper/demapper needed<br>• Limited functional splitting available<br>• Utilize the L2 switch supply chain<br>• Only Eth transport | • Open for performance enhancement<br>• Support any packet transport<br>• Flexible function splitting<br>• No support for CPRI | • Reuse the existing ODN<br>• Upgrade PON capacity and optimize the DBA algorithm |

# Thank you

www.huawei.com